

THE EVOLUTION OF AI · JULY 2026

FROM MODELS TO AGENTS

and the reinforcement learning turn

C O N T E N T S

- 01 the AI market today**
- 02 the models**
- 03 open source & post-training**
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Moonshot AI announcement sends US tech stocks tumbling

LAST YEAR

DeepSeek sparks AI stock selloff; Nvidia posts record market-cap loss

By Sinéad Carew, Amanda Cooper and Ankur Banerjee

January 27, 2025 6:00 PM CST · Updated January 27, 2025



02

Why

does this matter now?

\$582B

Corporate AI investment in 2025 — up ~130% in one year.

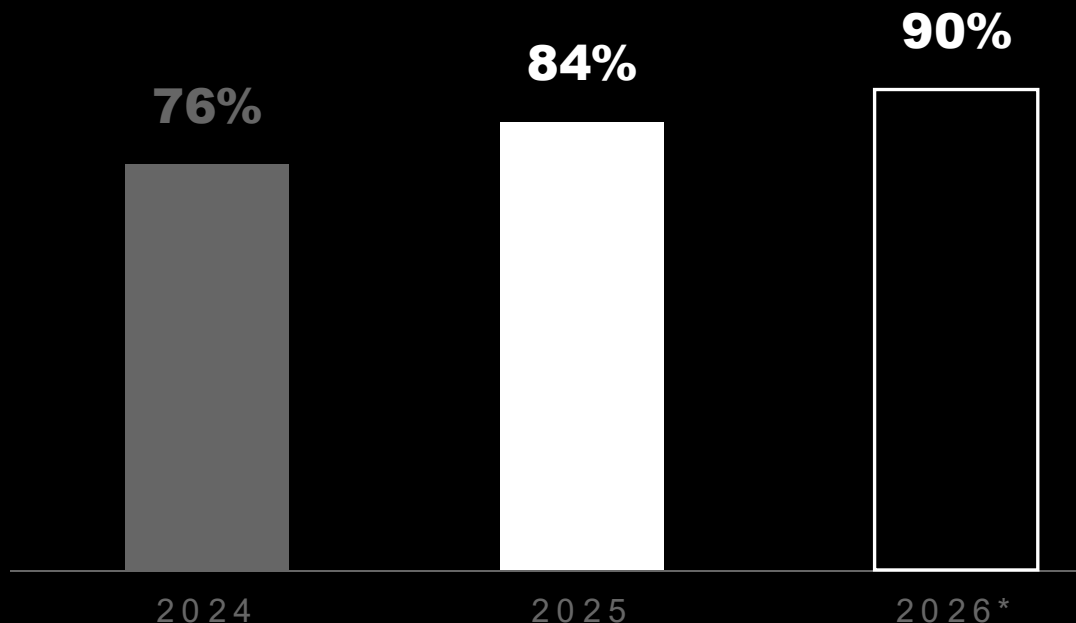
Stanford AI Index Report, 2026

53%

of the population adopted generative AI within three years.

Faster than the PC. Faster than the internet. — Stanford AI Index, 2026

84% of developers use or plan to use AI tools — and in 2026, 90% say they already use them at work.



*Stack Overflow Developer Survey 2024–25 · *JetBrains AI Pulse, Jan 2026 — % who regularly use AI at work*

01

How

did we get here?

2 0 1 2

machines learned to see.

AlexNet wins ImageNet — the deep learning era begins.

2 0 1 7

**attention is
all you need.**

The Transformer paper. Every modern model descends from it.

2020 — 2022

language at scale.

GPT-3, then Copilot GA, then ChatGPT — November 30, 2022.

2 0 2 4

models got hands.

Model Context Protocol, November 2024 — models connect to tools, data, the world.

T O D A Y

models became coworkers.

Agentic coding goes mainstream — heavy users run agents ~20 hours a week.

THE LAST SIX WEEKS

ANTHROPIC · JUNE 2026

Fable 5.

a Mythos-class model made safe for general use — its lead grows with task length.

OPENAI · JULY 2026

GPT-5.6 Sol.

flagship of the GPT-5.6 family — SOTA agentic benchmarks; 'ultra' runs four agents in parallel.

MOONSHOT · JULY 2026

Kimi K3.

2.8T open weights — tops agentic coding and browsing benchmarks.

every frontier release is now sold on the same promise: agents.

R L H F

**trained to
please.**

*human preferences as the reward —
InstructGPT, 2022.*

R L V R

**trained to be
right.**

*verifiable rewards — tests pass, answers check,
tasks complete.*

the reward moved from opinion to proof.

2024 — 2025

reasoning emerged from RL.

OpenAI o1, September 2024. DeepSeek-R1, January 2025 — reasoning from pure RL on verifiable rewards.

03

What

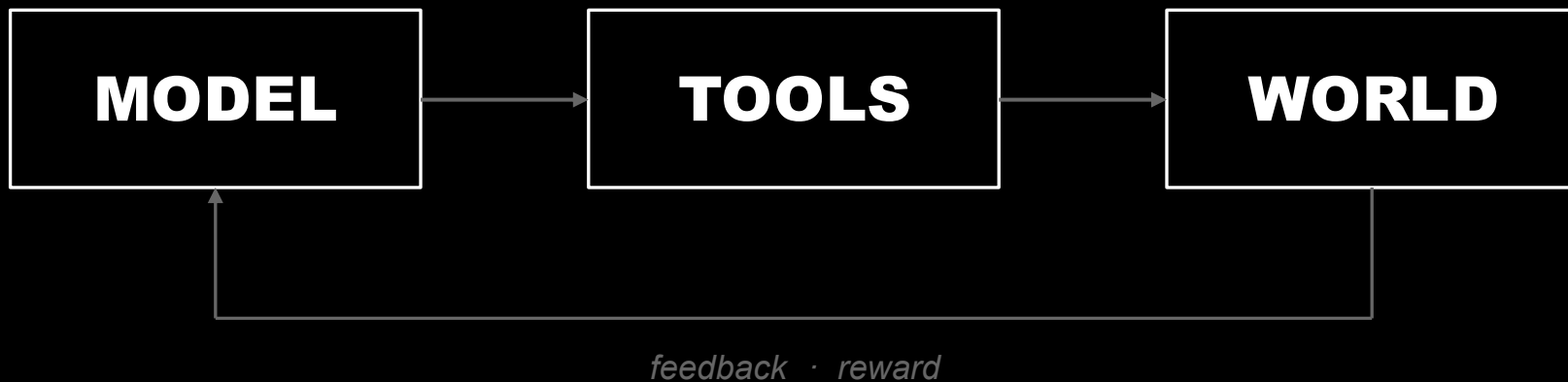
comes next?

an agent

**perceives. plans.
acts. verifies.**

a model in a loop — with tools, memory, and feedback.

the loop is the product.



2020

PREDICT →

complete the text

2022

CHAT →

answer the human

2026

ACT

finish the job

each era kept the last one's engine and changed the objective.

2022 → 2026

the craft moved up the stack.

From wording a request to designing the whole system the model runs in.

FOUR ERAS OF ENGINEERING IN ONE YEAR!!!!

PROMPT → **CONTEXT** → **HARNESS** → **LOOP**

choose the words

choose what it sees

choose its tools

choose how it learns

each layer engineers more of the model's world — the last one engineers the model itself.

2016

AlphaGo.

Reinforcement learning beats a world champion at Go — March 2016, Seoul.

reinforcement learning

try. score.

improve. repeat.

no labels, no lectures — just outcomes.

2026 →

agents that learn on the job.

RL environments for computer-use agents — 'training gyms' are now a funded frontier.

THE NEW BOTTLENECK

**the models are ready.
implementation
isn't**

the frontier ships monthly — getting AI deployed inside real companies is now the hard part.

RIGHT-SIZE THE MODEL

**you don't need a
frontier model
for everything.**

small, cheap, specialized models handle most production work — save the frontier for the hard problems.

THE FUTURE

every company will have one.

01 develop & post-train models

02 build & run agents

**03 track a frontier/models/tools that shifts
monthly**

like web teams in 2000 and data teams in 2015 — AI engineering becomes a standing function.

**“this is the
decade of agents.”**

ANDREJ KARPATHY · 2025

INTRODUCTION TO AI CLUB

Agentic AI @ UIUC

A student-run org on agentic AI — lectures build the vocabulary, then teams ship a real, deployed agent.

500

members in under two semesters.

courses teach concepts — this is where students practice.

THE CURRICULUM

- 01 prompting, LLMs, memory**
- 02 tools, function calling, structured output**
- 03 RAG & voice agents**
- 04 multi-agent orchestration & evaluation**

four lectures of vocabulary — then the repo is the exam.

8

deployed agent projects built by student teams.

equity, healthcare, voice, sports, food, music, portfolios, fabrication — github.com/agent-ai-uiuc

an agent that runs the lab.

Autonomous booking for the campus fab shop — a FIFO agent serves the next student, sends reminders, and an analytics chatbot answers staff questions.

Python · discord.py · FastAPI · React · SQLite — replaces paid SaaS for ~\$10/mo

call in. get booked. no human needed.

A voice agent that identifies the patient, triages symptoms by confidence score, finds open slots, and books, reschedules, or cancels — entirely over the phone.

Vapi.ai · FastAPI · Next.js · Supabase — tool-calling LLM over live phone audio

two products. one corpus.

A source-grounded medical Q&A bot for patients — and, from the same knowledge base, a simulator where med students interview an LLM patient and get scored.

LangGraph · Qdrant · Groq llama-3.3-70b · Flask — RAG state machine + agent layer

free food, found.

A searchable AI catalog of UIUC clubs that gathers org info, events, and free food from Instagram, Discord, Telegram, WhatsApp, and websites.

LangChain · ChromaDB · GPT-4o · Streamlit — multi-agent retrieval + valuation

AND FOUR MORE

sports analyst

equity agent

music rec

**portfolio
manager**

all public — github.com/agent-ai-uiuc

THE COMMON SHAPE

RETRIEVE → **DECIDE** → **ACT**

grounded in real data

an LLM picks the tool

a real side effect

every team built the same loop against a different domain.

SOCIAL MEDIA



CLUB-LINKEDIN



Discord



Personal LINKEDIN



A G E N T I C A I @ U I U C